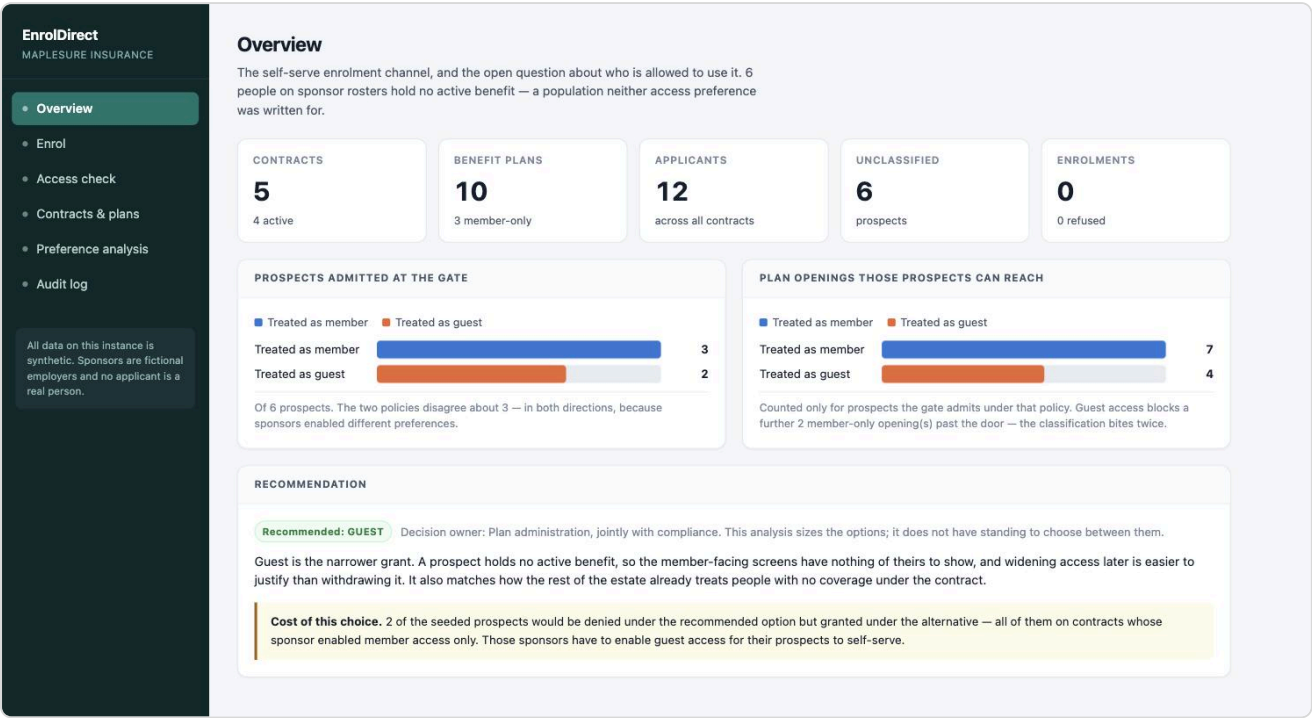


EnrolDirect

The online enrolment channel — what it is, what everything in it is called, how the access gate actually decides, and the user story we put through the AI pipeline on stage.

Screenshots captured live from `http://localhost:8083/` on 2026-08-04 · every figure in this document is the application's own computed output, not an illustration · all data synthetic (fictional sponsors, invented names)



What this document covers

1. **1** The application in one page
2. **2** The naming — every term, and which system owns it
3. **3** The logic — three gates, in an order that is a security property
4. **4** Screen by screen: Access check
5. **5** Screen by screen: Enrol
6. **6** Screen by screen: Contracts & plans
7. **7** Screen by screen: Preference analysis (4 tabs)
8. **8** Screen by screen: Audit log
9. **9** The user story — US-2026-045 / AMS-1045
10. **10** Before and after: the code the pipeline writes
11. **11** What proves it, and what does not

THE ONE-SENTENCE VERSION

EnrolDirect refuses a whole population today — not by a rule anyone wrote, but by an omission — and US-2026-045 is the change that settles it.

Overview. The whole problem on one screen: 12 applicants across 5 contracts, of whom **6 are prospects** — people on a sponsor's roster holding no active benefit, whom neither access preference was written for. Both charts and the recommendation are computed from the seeded directory on every request; nothing here is model output.

1 · The application in one page

EnrolDirect is MapleSure's self-serve enrolment channel. A plan member uses it to join or change benefits without calling the call centre. It is the only member-facing application of the three in this demo, which is why its refusal messages are customer-visible copy rather than internal diagnostics.

Enforcement and ownership are separate — and that is the whole point

Access is governed by two **access preferences** that the plan sponsor agrees at contract inception. Those preferences live on the contract record in **PolicyCore**. EnrolDirect *enforces* them; it does not own them.

WHY THE SPLIT IS LOAD-BEARING

Because PolicyCore owns the preferences, changing what a preference *means* is a change to a contract between two systems — not a local edit. That is exactly why the prospect question needed an impact analysis before anyone wrote a line of code, and why this app ships an analysis surface at all.

Runs on nothing but the venv

FastAPI + uvicorn, pinned. It seeds its own contracts and applicants in-process and calls no other service at runtime, so it starts alone and a locked-down sandbox can host it. Nothing survives a restart and nothing is supposed to.

```
apps/run-enroldirect.sh      # http://localhost:8083/
python -m pytest tests/test_regression_enroldirect.py
```

Port from `ENROLDIRECT_PORT` in `.env`, default 8083. A manager can also start, stop and reset it from the console's `/admin` panel without a terminal.

THE ESTATE, AS THE APP REPORTS IT

CONTRACTS

5

4 active, 1 lapsed

BENEFIT PLANS

10

3 member-only

APPLICANTS

12

across all contracts

TWO SURFACES, DELIBERATELY SEPARATED

SURFACE	ENDPOINTS	ANSWERS
The gate runtime; the only part a member touches	<code>/api/eligibility/*</code> <code>/api/enrolments</code> <code>/api/plans</code>	May this applicant use the channel, and what can they take once inside?
The analysis decision support; no member sees it	<code>/api/analysis/*</code>	Who consumes these preferences, how are they configured across the book, and what changes if prospects are classified one way or the other?

EVERY ANALYSIS FIGURE IS COMPUTED, NOT GENERATED

Pure functions in `impact.py` over the seeded directory. No model call, so no cache key, nothing to warm, and nothing that can be confidently wrong the way generated prose can. Same rule as the pipeline's own `diagram.py` and `acceptance.py`.

NOT THE SAME THING AS POLICYCORE'S ENROLMENT MODULE

`policycore/enrolment/` answers **when** may this person join — waiting periods, eligibility dates, life events, dependant age-out. `enroldirect/` answers whether the **online channel** is open to them at all. They compose; they do not overlap, and neither imports the other. Someone can be eligible to join and still have no online access — that is a correct outcome, not a bug.

2 · The naming — every term, and which system owns it

The two access preferences

Declared in `repos/enroldirect/preferences.py`

CONSTANT	STRING VALUE	WRITTEN FOR
MEMBER_ACCESS	"Online Enrolment – Member"	People already holding an active benefit under the contract, changing their own coverage.
GUEST_ACCESS	"Online Enrolment – Guest"	People with no active benefit who still have reason to enrol — retiree continuations, spousal transfers, sponsor-agreed exceptions.

THESE STRINGS ARE THE INTEGRATION CONTRACT, NOT DISPLAY LABELS

They arrive verbatim on the contract record from PolicyCore. An *unknown* preference is treated as *absent*, and absent means *not granted* — so renaming one here without renaming it there silently disables a gate rather than raising. `unknown_preferences()` reports the mismatch rather than throwing: one unrecognised string must not take the whole channel down.

The three applicant categories

Declared in `repos/enroldirect/applicants.py`. Category is derived *upstream* by plan administration — EnrolDirect is told what someone is, it does not decide.

CONSTANT	PEOPLE	ON ROSTER?	ACTIVE BENEFIT?	HAS A PREFERENCE?
MEMBER	4	yes	yes	YES — MEMBER
GUEST	2	no	no	YES — GUEST
PROSPECT	6	yes	no	NONE

A prospect is not a halfway state on the way to becoming a member — it is a population the sponsor has already accepted onto the roster who has not taken up coverage. Treating them as a *guest* (a stranger to the contract) is uncomfortable; treating them as a *member* (someone with coverage to change) is inaccurate.

The decision object

`EligibilityDecision` in `eligibility.py` — returned by `POST /api/eligibility/check`

FIELD	MEANING
granted	Did every gate open.
category	The applicant's own category, as received.
requiredPreference	What was <i>consulted</i> for this applicant, enabled or not. <code>None</code> when the category resolved to no preference at all.
authorisingPreference	What actually <i>opened</i> the gate. <code>None</code> on every denial.
reasons	Why, in customer-visible English. Never a bare boolean.

WHY TWO PREFERENCE FIELDS THAT LOOK THE SAME

Today they coincide for everyone admitted, because only members and guests are admitted at all. They are still kept apart, because DocumentHub words the confirmation pack from `authorisingPreference` and NightlyBatch reconciles on it. The moment those two diverge — which is exactly what US-2026-045 causes — a single field would be silently wrong somewhere downstream.

Benefit plans and tiers

TERM	MEANING
planCode	PL-1001 ... — one plan under one group contract.
memberOnly	The plan attaches to existing coverage, so there is nothing for it to attach to for someone holding none. 3 of the 10 seeded plans.
COVERAGE_TIERS	Single, Couple, Family — cheapest first; the order is load-bearing. Not every plan is sold at every tier, and an absent tier is real configuration, not a lookup failure.

Identifiers used on stage

US-2026-045 is the user story file, `stories/US-2026-045.md`; AMS-1045 is its ticket key on the board, derived from the number rather than the prefix. `enroldirect-prospect-access` is the S3 target id and `enroldirect_prospect_access` its replay cache namespace — never rename that one, every committed recording resolves through it. Seeded data: MS-2001 ... MS-2005 are the five group contracts, AP-4001 ... AP-4012 the twelve applicants.

3 · The logic — three gates, in an order that is a security property

POST /api/eligibility/check runs three gates. All three have to open, and the order is fixed.

- 1

The contract is ACTIVE

A lapsed contract keeps whatever preferences it was configured with, so this runs first. Read preferences off a lapsed contract and stale configuration grants access to a live channel.
- 2

The applicant's category resolves to a preference

Members → Member access. Guests → Guest access. **Prospects → nothing.** `preference_for_category()` returns `None`, and that is where a prospect is refused today.
- 3

The sponsor enabled that preference

The sponsor agreed this at inception. Refusal here is `NO_CHANNEL_ACCESS` — refused at the door, not at the catalogue.

REVERSING GATES 1 AND 3 IS INVISIBLE TO CATEGORY-LEVEL TESTING

Every category would still resolve to the right preference; every preference check would still work. The only broken case is a lapsed contract that still has the preference enabled — which is precisely the case nobody thinks to write a test for. `MS-2005` exists in the seed for this reason, and both the unit and regression suites pin the ordering against it explicitly.

Enrolment reuses the gate — it does not reimplement it

POST /api/enrolments runs four checks: the access gate above (reused), then the plan is on that applicant's contract, then the plan is open to their effective category, then the plan is sold at the requested tier.

An enrolment path carrying its own copy of the access rules is how a channel ends up admitting someone the gate would have turned away. The two copies drift on the first change to either.

A REFUSAL IS A 200, NOT A 4XX

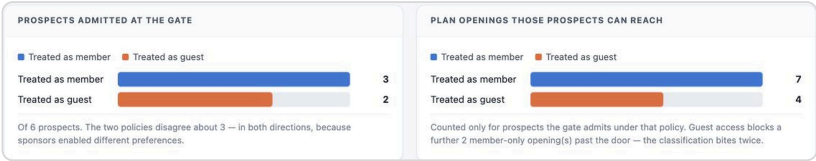
The applicant asked a valid question and got a valid, recorded answer. Turning "you may not enrol in this plan" into an HTTP error would lose the audit entry that says so and why. Refusals are kept, not discarded: "how many failed" is a number nobody can act on; "how many failed because the plan needed existing coverage" changes a decision.

The classification bites twice

Because some plans are `memberOnly`, the prospect question decides **two** things, and the two effects run in *opposite* directions.

	WHAT IT DECIDES	WHICH OPTION REACHES FURTHER
At the gate	How many prospects are admitted at all	MEMBER admits more
At the catalogue	How much of the catalogue those admitted can reach	MEMBER reaches more

LIVE OUTPUT · OVERVIEW



Both effects, measured separately. Catalogue reach is counted *only* for prospects the gate would admit under that option — a plan you cannot reach because you were refused at the door is already counted as a denial, and counting it twice would inflate the gap.

Treated as member

Treated as guest

 Colour follows the policy in every figure; every bar carries its value.

Data faults fail at the boundary

`Applicant.__post_init__` rejects two combinations outright rather than coercing them to the "obviously intended" category: **MEMBER with no active benefit** (that is a prospect, mislabelled upstream) and **PROSPECT holding an active benefit** (that is a member, not yet promoted).

Both would grant or deny the wrong access *silently*. Coercing them hides an upstream data quality problem inside the system that consumes it — and the consuming system is the one that gets blamed for the wrong outcome. Failing loudly at construction puts the error next to the bad record.

4 · Screen by screen — Access check

The gate, exposed directly: pick a seeded applicant, run the three gates, read the decision and its reasons. This is the screen that shows the defect in one click.

A MEMBER — THE GATE OPENS

EnrolDirect

MAPLEBURE INSURANCE

• Overview

• Enrol

• Access check

• Contracts & plans

• Preference analysis

• Audit log

All data on this instance is synthetic. Sponsors are fictional employees and no applicant is a real person.

Access check

Three gates, in order: the contract is active, the applicant's category resolves to a preference, and the sponsor enabled that preference.

RUN A CHECK

Applicant

Rowan Iqbal — MEMBER · MS-2001

Check access

Every decision carries its reasons. A denial that cannot say which gate closed becomes a support ticket.

DECISION

GRANTED Rowan Iqbal

ContractMS-2001 — Northwind Logistics Ltd.

CategoryMEMBER

Preference requiredOnline Enrolment - Member

Authorised byOnline Enrolment - Member

• Granted under 'Online Enrolment - Member'.

DECISION

GRANTED

Rowan Iqbal

ContractMS-2001 — Northwind Logistics Ltd.

CategoryMEMBER

Preference requiredOnline Enrolment - Member

Authorised byOnline Enrolment - Member

• Granted under 'Online Enrolment - Member'.

AP-4001 Rowan Iqbal, MEMBER on MS-2001. Category resolves to Member access, the sponsor enabled it, and `authorisingPreference` records which preference did the work.

A PROSPECT — THE GATE CLOSES, AND THIS IS THE DEFECT

EnrolDirect

MAPLEBURE INSURANCE

• Overview

• Enrol

• Access check

• Contracts & plans

• Preference analysis

• Audit log

All data on this instance is synthetic. Sponsors are fictional employees and no applicant is a real person.

Access check

Three gates, in order: the contract is active, the applicant's category resolves to a preference, and the sponsor enabled that preference.

RUN A CHECK

Applicant

Marguerite Vasseur — PROSPECT · MS-2002

Check access

Every decision carries its reasons. A denial that cannot say which gate closed becomes a support ticket.

DECISION

DENIED Marguerite Vasseur

ContractMS-2002 — Cedarline Manufacturing Inc.

CategoryPROSPECT

Preference required—

Authorised by—

• Applicant category PROSPECT has no online enrolment preference and cannot be granted access.

DECISION

DENIED

Marguerite Vasseur

ContractMS-2002 — Cedarline Manufacturing Inc.

CategoryPROSPECT

Preference required—

Authorised by—

• Applicant category PROSPECT has no online enrolment preference and cannot be granted access.

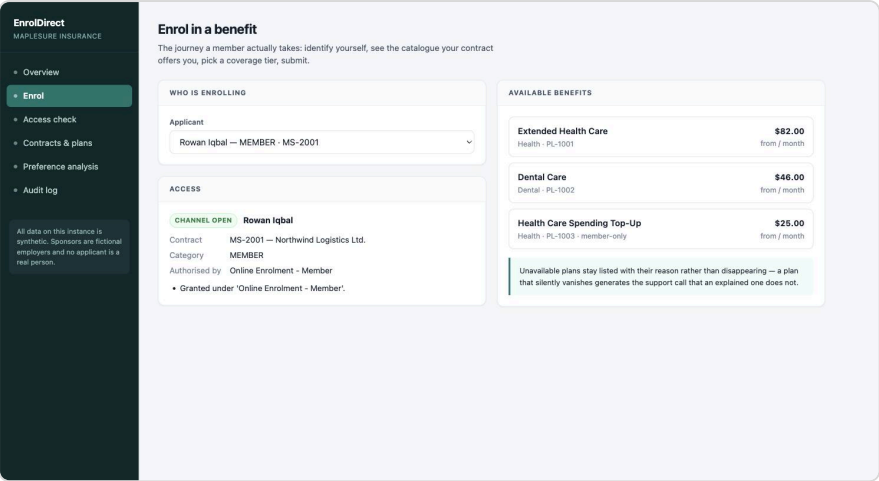
AP-4005 Marguerite Vasseur, PROSPECT on MS-2002. She is on Cedarline's roster. The contract is active. The sponsor *has* enabled online enrolment. She is refused at gate 2 — `requiredPreference` is *empty*, because no preference resolves for her category at all.

The line to say out loud: "That refusal is not a decision anyone took. It is the narrower option in force by default — and every day it stands is a day the sponsor's own rostered people are turned away by an omission rather than by a rule."

5 · Screen by screen — Enrol

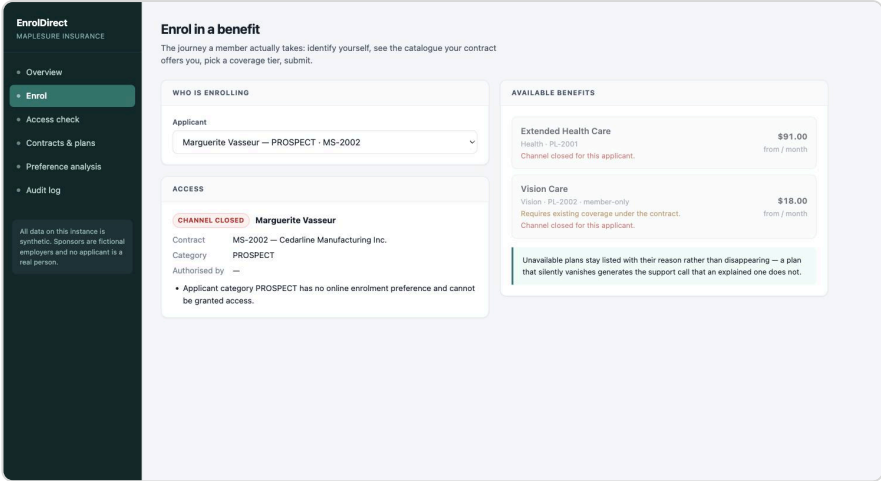
The journey a member actually takes: identify yourself, see the catalogue your contract offers *you*, pick a tier, submit. The access panel on the left is the same gate as the previous screen — reused, not reimplemented.

CHANNEL OPEN

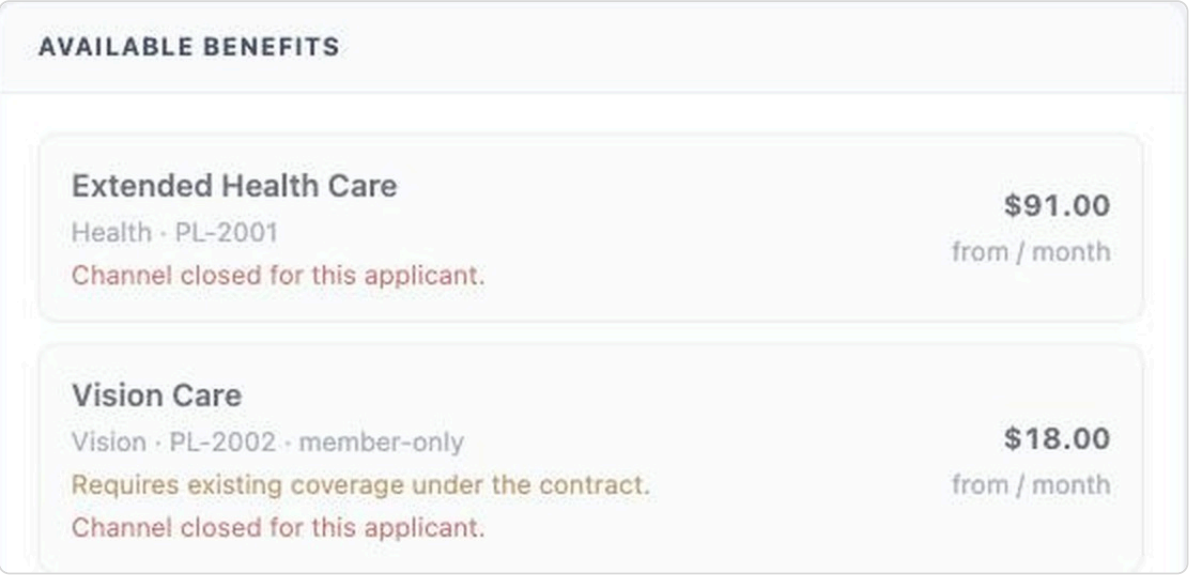


A member sees three plans, priced. Including PL-1003, a member-only top-up — it attaches to the health benefit they already hold.

CHANNEL CLOSED



A prospect sees the catalogue and cannot touch any of it. Note the two distinct reasons stacked on Vision Care — one from the gate, one from the plan.



Unavailable plans stay listed, with their reason. A plan that silently vanishes generates the support call that an explained one does not — "why can't I see dental?" has no answer in a log.

Why both reasons appear, separately

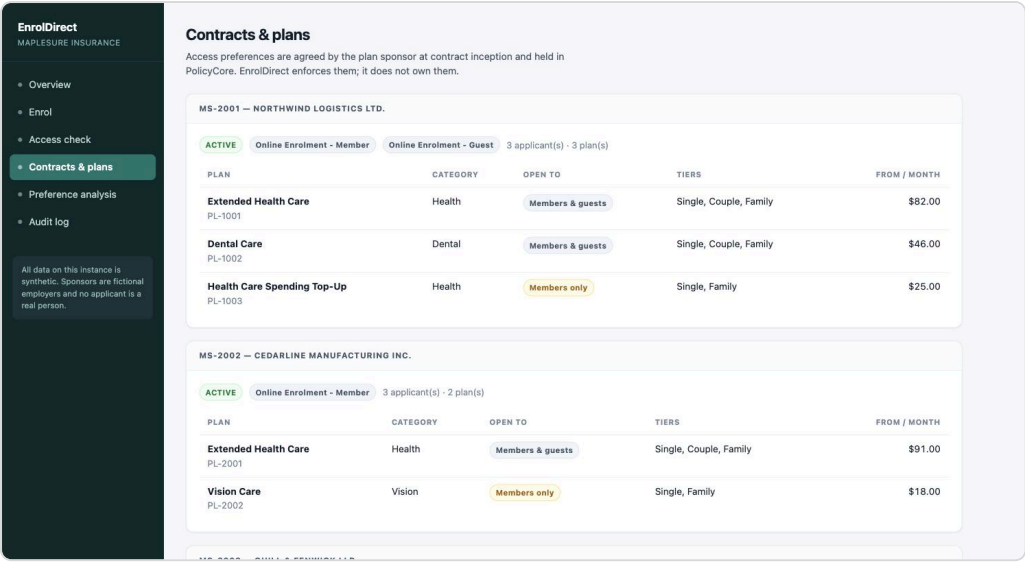
CHANNEL CLOSED FOR THIS APPLICANT comes from the **gate** — it applies to every plan on the contract.

REQUIRES EXISTING COVERAGE comes from the **plan** — memberOnly, and it would apply to a guest who *had* been admitted.

Keeping them apart makes the two-bite effect visible in the product, not just in the analysis: once US-2026-045 admits this prospect as a guest, the first line disappears and the second one stays.

A decision without a reason is a defect. This is member-facing — the refusal text is the product for anyone refused, and a denial that cannot name the gate becomes a phone call.

6 · Screen by screen — Contracts & plans



The configuration the gate reads — each contract's status, the preferences the sponsor enabled, and its plans with *open to* and tier pricing. PolicyCore's data, rendered by the system that enforces it.

The full catalogue — 10 plans, 3 of them member-only

The highlighted rows are what makes the classification bite twice. Premiums are synthetic monthly CAD amounts, not a rate table. Lapsed MS-2005's one plan is omitted — gate 1 closes before anyone reaches its catalogue.

PLAN	CONTRACT	CATEGORY	OPEN TO	TIERS	FROM
PL-1001	MS-2001	Health	ALL	Single / Couple / Family	\$82
PL-1002	MS-2001	Dental	ALL	Single / Couple / Family	\$46
PL-1003	MS-2001	Health	MEMBER-ONLY	Single / Family	\$25
PL-2001	MS-2002	Health	ALL	Single / Couple / Family	\$91
PL-2002	MS-2002	Vision	MEMBER-ONLY	Single / Family	\$18
PL-3001	MS-2003	Health	ALL	Single / Couple / Family	\$77
PL-3002	MS-2003	Life	ALL	Single	\$12
PL-3003	MS-2003	Disability	MEMBER-ONLY	Single	\$34
PL-4001	MS-2004	Health	ALL	Single / Couple / Family	\$88

The seed is shaped deliberately

Every combination of enabled preferences appears at least once, and prospects sit on contracts whose preferences point in *opposite* directions — so the comparison has genuine disagreement to report. A seed where both options produced the same answer everywhere would make the analysis look decisive and prove nothing.

CONTRACT	STATUS	ENABLED	WHAT IT PROVES
MS-2001 Northwind	ACTIVE	Member + Guest	Its prospect is granted either way — which makes the two options look equivalent until you look further.
MS-2002 Cedarline	ACTIVE	Member only	Its 2 prospects are granted under MEMBER and denied under GUEST.
MS-2003 Quill & Fenwick	ACTIVE	Guest only	The disagreement runs the <i>other</i> way here.
MS-2004 Talus	ACTIVE	none	Nobody is granted at all — proof the analysis reads contract configuration, not category alone. An empty list is real configuration, not missing data.
MS-2005 Harbourline	LAPSED	Member + Guest	The gate-ordering case. Both preferences still configured; everyone must still be denied.

EnrolDirect

MAPLESURE INSURANCE

• Overview

• Enrol

• Access check

• Contracts & plans

• Preference analysis

• Audit log

All data on this instance is synthetic. Sponsors are fictional employers and no applicant is a real person.

Preference analysis

Which systems consume the access preferences, how they are configured across the book, and what changes if prospects are classified one way or the other. Every figure is computed from the directory — no model call.

Prospect impact

Catalogue reach

Preference usage

Consumers

POLICY	REQUIRES PREFERENCE	GRANTED	DENIED
MEMBER	Online Enrolment - Member	3	3
GUESTrecommended	Online Enrolment - Guest	2	4

WHERE THE TWO POLICIES DISAGREE

APPLICANT	CONTRACT	SPONSOR ENABLED	GRANTED ONLY BY
AP-4005	MS-2002 — Cedarline Manufacturing Inc.	Online Enrolment - Member	MEMBER
AP-4006	MS-2002 — Cedarline Manufacturing Inc.	Online Enrolment - Member	MEMBER
AP-4008	MS-2003 — Quill & Fenwick LLP	Online Enrolment - Guest	GUEST

Recommendation: GUEST.

Guest is the narrower grant. A prospect holds no active benefit, so the member-facing screens have nothing of theirs to show, and widening access later is easier to justify than withdrawing it. It also matches how the rest of the estate already treats people with no coverage under the contract.

Cost of this choice.

2 of the seeded prospects would be denied under the recommended option but granted under the alternative — all of them on contracts whose sponsor enabled member access only. Those sponsors have to enable guest access for their prospects to self-serve.

Decision owner.

Plan administration, jointly with compliance. This analysis sizes the options; it does not have standing to choose between them.

NOT EVIDENCED BY THIS ANALYSIS

• Regulatory position.

Whether a prospect admitted under member access has been represented as a plan member is a compliance question, and no count of contracts

Both options priced, not just the recommended one. Pricing only the option you advocate is advocacy. The disagreement table names every applicant the two options decide differently — and it runs in both directions, which is what stops the comparison from being a one-sided argument.

The recommendation, and its cost

RECOMMENDED

GUEST

the narrower grant

DISAGREEMENTS

3

of 6 prospects

Why Guest. A prospect holds no active benefit, so the member-facing screens have nothing of theirs to show; widening access later is easier to justify than withdrawing it; and it matches how the rest of the estate already treats people with no coverage under a contract.

THE RECOMMENDATION REPORTS ITS OWN COST

2 of the seeded prospects would be denied under the recommended option but granted under the alternative — all of them on contracts whose sponsor enabled member access only. Those sponsors have to enable guest access for their prospects to self-serve.

Decision owner: plan administration, jointly with compliance. The analysis sizes the options; it does not have standing to choose between them. `implemented: false` until the story runs.

7 · Preference analysis (tabs 2 and 3) — the second bite, and how the book is configured

TAB 2 · CATALOGUE REACH

EnrolDirect

MAPLESURE INSURANCE

Overview

Enrol

Access check

Contracts & plans

Preference analysis

Audit log

All data on this instance is synthetic. Sponsors are fictional employers and no applicant is a real person.

Preference analysis

Which systems consume the access preferences, how they are configured across the book, and what changes if prospects are classified one way or the other. Every figure is computed from the directory — no model call.

Prospect impactCatalogue reachPreference usageConsumers

Counted only for prospects the gate would admit under that option — a plan you cannot reach because you were refused at the door is already counted as a denial.

POLICY	PROSPECTS ADMITTED	PLAN OPENINGS REACHABLE	BLOCKED AS MEMBER-ONLY
MEMBER	3	7	0
QUEST	2	4	2

The classification bites twice. Guest access admits fewer prospects at the gate and gives the ones it admits a smaller catalogue, because member-only plans attach to coverage a prospect does not hold.

The second bite, measured on its own: Guest access admits fewer prospects *and* gives the ones it admits a smaller catalogue — 4 plan openings against 7, with 2 blocked as member-only.

TAB 3 · PREFERENCE USAGE

EnrolDirect

MAPLESURE INSURANCE

Overview

Enrol

Access check

Contracts & plans

Preference analysis

Audit log

All data on this instance is synthetic. Sponsors are fictional employers and no applicant is a real person.

Preference analysis

Which systems consume the access preferences, how they are configured across the book, and what changes if prospects are classified one way or the other. Every figure is computed from the directory — no model call.

Prospect impactCatalogue reachPreference usageConsumers

PREFERENCE	CONTRACTS	ENABLED ON
Online Enrolment - Member Plan members already enrolled in at least one benefit under the contract, using the self-serve channel to change their own coverage.	3	MS-2001, MS-2002, MS-2005
Online Enrolment - Guest People with no active benefit under the contract who still have a reason to enrol — retiree continuations, spousal transfers, and other sponsor-agreed exceptions.	3	MS-2001, MS-2003, MS-2005

APPLICANT POPULATION	PEOPLE	HAS A PREFERENCE
MEMBER On the contract roster, holding at least one active benefit.	4	yes
GUEST Not on the contract roster. Enrolling under a sponsor-agreed exception.	2	yes
PROSPECT On the contract roster, holding no active benefit yet.	6	no

How the book is actually configured — 3 contracts enable each preference — and the population per category. **PROSPECT: 6 people, no preference.**

EnrolDirect

MAPLESURE INSURANCE

Overview

Enrol

Access check

Contracts & plans

Preference analysis

Audit log

All data on this instance is synthetic. Sponsors are fictional employers and no applicant is a real person.

AP-4005	MS-2002 — Cedarline Manufacturing Inc.	Online Enrolment - Member	MEMBER
AP-4006	MS-2002 — Cedarline Manufacturing Inc.	Online Enrolment - Member	MEMBER
AP-4008	MS-2003 — Quill & Ferwick LLP	Online Enrolment - Guest	QUEST

Recommendation: QUEST. Guest is the narrower grant. A prospect holds no active benefit, so the member-facing screens have nothing of theirs to show, and widening access later is easier to justify than withdrawing it. It also matches how the rest of the estate already treats people with no coverage under the contract.

Cost of this choice. 2 of the seeded prospects would be denied under the recommended option but granted under the alternative — all of them on contracts whose sponsor enabled member access only. Those sponsors have to enable guest access for their prospects to self-serve.

Decision owner. Plan administration, jointly with compliance. This analysis sizes the options; it does not have standing to choose between them.

NOT EVIDENCED BY THIS ANALYSIS

- Regulatory position. Whether a prospect admitted under member access has been represented as a plan member is a compliance question, and no count of contracts answers it.
- Support load. Denied prospects contact the service desk; admitted prospects reaching member screens with no coverage to display also contact the service desk. The seed cannot say which is the larger volume.
- Downstream subscriber behaviour. The consumers above are a declared list, not a discovered one. A subscriber filtering on the authorising preference that nobody recorded here will change behaviour unannounced.
- Prospect expectation. What the sponsor told these people they would be able to do is not in the contract record.

STANDING RISKS

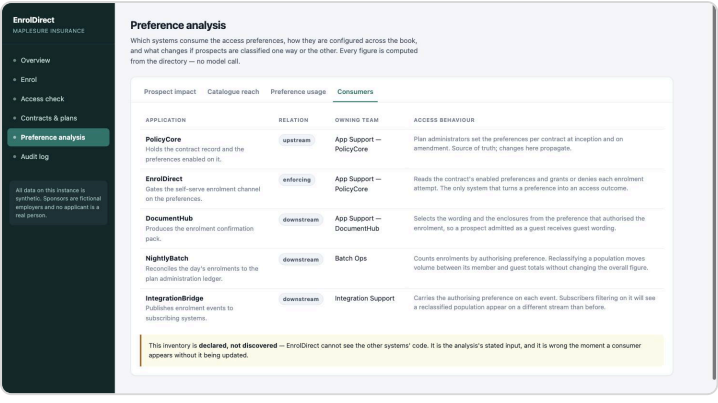
- A contract carrying a preference this app does not recognise is treated as not having it, which denies rather than grants — safe, but silent. See 'preferences.unknown_preferences'.
- The contract gate runs before the preference gate, so a lapsed contract's stale preferences cannot grant access. Reordering those two gates would reintroduce that hole without failing any category-level test.
- Prospects are refused at the gate today because no preference resolves for them. That is the current behaviour, not a decision — every day it stands is a day the narrower option is in force by default rather than by choice.
- Whichever option is adopted would be a single value for the whole book. Any per-sponsor variation would need it to become contract data, which is a PolicyCore change, not an EnrolDirect one.

"Not evidenced by this analysis" is load-bearing. Regulatory position, support load, undeclared subscribers and what a prospect was told they could do are real consequences configuration data cannot size. They are named rather than dropped — an analysis listing only supporting numbers is advocacy, the same rule as the release record's `unproven_claims()`.

THE THIRD STANDING RISK IS THE ONE TO READ ALOUD

"Prospects are refused at the gate today because no preference resolves for them. That is the current behaviour, not a decision — every day it stands is a day the narrower option is in force by default."

The analysis recomputes from current configuration on every request rather than serving a stored conclusion — an analysis *surface*, not a document. It cannot go stale relative to the rules it mirrors, and the regression suite pins it to them so the two cannot drift.



Five systems read or write these preferences — one upstream owner, one enforcer, three downstream consumers, each with its owning team named so a ticket has somewhere to go. The inventory is **declared, not discovered**: this app cannot see the other systems' code, so it is wrong the moment a consumer appears without the table being updated.

Being affected is not the same as having work

All five stay in the inventory — the estate map is not the ticket list. Only **one** of them has to author behaviour it has never had:

SYSTEM	CODE CHANGE?	WHY
DocumentHub	REQUIRED	It words the confirmation pack from the authorising preference alone. A rostered person admitted under guest access is a recipient neither existing pack was written for.
NightlyBatch	NO	Aggregates by authorising preference into buckets it already has. Its numbers move; its code does not.
IntegrationBridge	NO	Carries the preference through on an existing field.

THE BAR IS "THAT TEAM MUST CHANGE CODE", NOT "THAT TEAM IS AFFECTED"

The check used to return every system the change touched — three tickets, two of them fictional, and nobody notices they are fictional until three teams have triaged them. A "no" is now recorded with its rationale (`changeRequired + changeRationale`), auditable rather than an omission.

The downstream consequence is sized, not described

IF PROSPECTS ARE...	PACKS NEEDING NEW WORDING	CONSEQUENCE IF DOCUMENTHUB IS UNCHANGED
MEMBERS	3	The member pack refers to coverage the recipient already holds. A prospect holds none, so the paragraph describes benefits that do not exist.
GUESTS (rec.)	2	The guest pack states no member record is held and encloses an identity form. Every recipient is on the sponsor's roster, so both are false and the form has to be withdrawn by hand.

Computed by `impact.document_impact()` — no LLM. Both options are priced, not just the recommended one; pricing only the option you advocate is advocacy. It rides inside `prospect_impact()["documentImpact"]` rather than on an endpoint of its own, because a downstream consequence served separately is one a reader can finish the analysis without ever having seen.

Why this matters to the demo

DocumentHub is S3's fourth target and the only one registered by a dropped-in `.s3targets.json` manifest rather than declared by hand — the cross-team check named it, the repo was dropped into `repos/`, and it became automatable with no edit to the target list.

Its own baseline is a **wrong document, not a failure**: the seeded record `ENR-20260804-005` — on the roster, admitted under guest access, exactly what US-2026-045 starts producing — falls through to the guest pack and is told we hold no member record for them. Nothing raises and nothing logs.

8 · Screen by screen — Audit log

EnrolDirect

MAPLESURE INSURANCE

• Overview

• Enrol

• Access check

• Contracts & plans

• Preference analysis

• Audit log

All data on this instance is synthetic. Sponsors are fictional employers and no applicant is a real person.

Audit log

Every enrolment attempt, successful or refused, with the preference that authorised it.

ATTEMPTS

4

ENROLLED

3

REFUSED

1

MONTHLY PREMIUM

\$285.00

enrolled total

ATTEMPTS — NEWEST FIRST

Clear log

In-process only; nothing persists across a restart.

REFERENCE	APPLICANT	PLAN	STATUS	AUTHORISED BY	SUBMITTED
ENR-000004	Rowan Iqbal MEMBER	Health Care Spending Top-Up Single	SUBMITTED	Online Enrolment - Member	2026-08-04T06:52:32+00:00
ENR-000003	Marguerite Vasseur PROSPECT	Extended Health Care	REFUSED NO_CHANNEL_ACCESS	—	2026-08-04T06:52:32+00:00
ENR-000002	Priya Chandrasekar GUEST	Dental Care Single	SUBMITTED	Online Enrolment - Guest	2026-08-04T06:52:32+00:00
ENR-000001	Rowan Iqbal MEMBER	Extended Health Care Family	SUBMITTED	Online Enrolment - Member	2026-08-04T06:52:32+00:00

WHY ATTEMPTS WERE REFUSED

REASON	COUNT
NO_CHANNEL_ACCESS	1

Refusals are recorded, not discarded. "How many failed" is a number nobody can act on; "how many failed because the plan needed existing coverage" changes a decision.

Every attempt, successful or refused, with the preference that authorised it. Shown here after four sample submissions — the refused prospect (ENR-000003) carries NO_CHANNEL_ACCESS and no authorising preference.

ATTEMPTS

4

ENROLLED

3

REFUSED

1

MONTHLY PREMIUM

\$285.00

enrolled total

Refusals are counted, not just successes.

Refusal codes

CODE	REFUSED AT
NO_CHANNEL_ACCESS	The gate — the channel is closed to this applicant.
PLAN_NOT_ON_CONTRACT	The plan is not offered under their contract.
MEMBER_ONLY_PLAN	The catalogue — the plan needs existing coverage.
TIER_NOT_OFFERED	The plan is not sold at the requested tier.

AC2 of the user story pins these: every existing refusal code keeps its current meaning, and a prospect refused because their sponsor did not enable the resolved preference is refused at the gate (NO_CHANNEL_ACCESS), not at the catalogue.

THE COST OF NO PERSISTENCE, STATED RATHER THAN GLOSSED

The log is in-process and dies with the process (POST /api/enrolments/reset clears it for a second rehearsal).

The sandbox constraint means a datastore is a dependency this app cannot take — but that makes the enrolment log **unsuitable as the audit record of last resort**. Access decisions are the evidence trail if a refusal is disputed. That is a gap in the current design, not a feature.

9 · The user story we run on stage — US-2026-045 / AMS-1045

STORY

US-2026-045

stories/US-2026-045.md

TICKET

AMS-1045

issue type: Story

PRIORITY

P3

enhancement

Title: **Prospect Member Eligibility Check For Online Enrolment** · Requested by MapleSure Group Retirement Product
· Application: EnrolDirect

Business objective

Determine whether a prospect member is eligible for online enrolment, and return that determination through the channel's *existing* eligibility check — so the systems downstream can act on a real answer instead of a refusal.

Target member type

A member who is on the plan sponsor's contract roster, has not enrolled in any plan, and holds no active benefit. EnrolDirect already has a name for this population — **PROSPECT** — and already distinguishes it from the two the access preferences were written for.

Acceptance criteria

AC	GIVEN / WHEN / THEN
AC1 ELIGIBLE	Given a member on the contract roster holding no active benefit, when they pass every gate, then the check reports them eligible and the decision <i>names the preference that admitted them</i> .
AC2 NOT ELIGIBLE	Given the same member, when they fail any one gate, then the check reports them not eligible and the decision carries the reason. Every existing refusal code keeps its current meaning.
AC3 the rules	The determination is built from the eight named rules opposite — the story states them, so the model is not left to invent an architecture.

IT IS A "USER STORY" EVERYWHERE — "CR" IS GONE

The client's own intake artifact is a user story (business objective, target population, Given/When/Then), so the vocabulary follows it end to end: `stories/`, `US-YYYY-NNN`, `/api/s3/story`. The board opens it as issue type **Story** with no id prefix on the card.

The classification decides two things, and both are in scope

At the gate it decides how many prospects are admitted. Past the gate it decides how much of the benefit catalogue those admitted can reach, because member-only plans attach to existing coverage. **The two effects run in opposite directions, and the change is not complete if only the first is made.**

Explicitly out of scope

NOT CHANGED	WHY
A third access preference	Would change what PolicyCore stores and what every consumer reads.
Per-contract variation	Same — it would have to become contract data, a PolicyCore change.
impact.py	The analysis is the <i>input</i> to this change, not part of it. It must keep sizing both options after one is adopted — a comparison reporting only the option in force could not answer "what is this costing us", which is the question that reopens the decision.
static/index.html	The console must keep rendering with no edits.
tests/test_regression_enrolldirect.py	The independent check that the story broke nothing. Must pass unchanged, before and after.

9 · US-2026-045 — AC3, the eight rules the determination is built from

The story does not just ask for an outcome; it names the shape. This is the part that makes the run reproducible — the model is not left to invent an architecture, and each rule is checkable in the diff.

The rules

1. The prospect policy is a **single named value** in `applicants.py`, set to treat prospects as guests, sitting alongside the two options it selects between. Module-level configuration, *not* a per-call parameter — a caller that could pass its own would be choosing its own access rules.
2. `preference_for_category` resolves a prospect through that policy. Members and guests keep resolving exactly as they do now, and it stays **the single place** a category becomes a preference.
3. A new `effective_category` returns the category an applicant is *treated as* for the rest of the journey. Resolved once and reused, not re-derived at each call site.
4. The three gates keep their current order. A lapsed contract must still deny a prospect regardless of the policy.
5. The decision records *which policy* resolved the preference — populated for prospects, null for everyone else. `authorisingPreference` keeps meaning the preference that actually opened the gate.
6. The enrolment path applies the same effective category and **reuses the access gate** rather than reimplementing it.
7. The benefit catalogue endpoint reports and filters on the effective category, so the catalogue and the gate cannot disagree about the same person.
8. **Members and guests are unaffected**, and the regression suite keeps passing without changes to it.

Why each of these is worded the way it is

RULE	THE FAILURE IT PREVENTS
1 · module-level, not per-call	A caller that could pass its own policy would be choosing its own access rules. The gate enforces one policy.
2 · one lookup, one place	<code>preference_for_category</code> was written as a single lookup rather than a branch repeated at each call site <i>specifically</i> so that whatever was eventually decided would land in exactly one function.
3 · resolved once, reused	Deriving the effective category again at each call site is how the catalogue and the gate end up disagreeing about the same person.
4 · gate order unchanged	A lapsed contract must still deny a prospect. The policy must not become a way past gate 1.
5 · <code>authorisingPreference</code> keeps its meaning	It is what DocumentHub words the pack from and what NightlyBatch reconciles on. A prospect admitted as a guest must be recorded as such.
6 · reuse the gate	Two copies of the access rules drift on the first change to either.
7 · catalogue filters on the same value	The second bite. Make only the first change and the story is half-done.
8 · members and guests unaffected	If the change moved a member's outcome it would not be contained to the population it was scoped to.

10 · Before and after — the code the pipeline writes

The one function that changes meaning

repos/enroldirect/eligibility.py — before

```
def preference_for_category(category: str) -> str | None:
    if category == MEMBER:
        return MEMBER_ACCESS
    if category == GUEST:
        return GUEST_ACCESS
    return None          # ← a prospect is refused here
```

after

```
def preference_for_category(category: str) -> str | None:
    if category == MEMBER:
        return MEMBER_ACCESS
    if category == GUEST:
        return GUEST_ACCESS
    if category == PROSPECT:
        return (MEMBER_ACCESS if PROSPECT_POLICY == TREAT_AS_MEMBER
                else GUEST_ACCESS)
    return None
```

and the policy it reads, in applicants.py

```
# The two policies available for prospects.
TREAT_AS_MEMBER = "MEMBER"
TREAT_AS_GUEST = "GUEST"

PROSPECT_POLICIES: tuple[str, ...] = (TREAT_AS_MEMBER, TREAT_AS_GUEST)

# The current policy in force for prospects.
PROSPECT_POLICY = TREAT_AS_GUEST
```

and the new resolver plus the new decision field

```
def effective_category(category: str) -> str:
    """The category an applicant is treated as onward."""
    if category == PROSPECT:
        return PROSPECT_POLICY
    return category

@dataclass(frozen=True)
class EligibilityDecision:
    ...
    authorisingPreference: str | None
    prospectPolicyApplied: str | None = None
```

What the model may and may not touch

The target declares six core files. Only four of them are editable — and that asymmetry is the point of this target.

FILE	IN PROMPT	EDITABLE
applicants.py	YES	YES
eligibility.py	YES	YES
enrolments.py	YES	YES
main.py	YES	YES
impact.py	YES	NO
preferences.py	YES	NO

READ-BUT-DON'T-EDIT IS ENFORCED, NOT REQUESTED

The model *has* to read `impact.py` to understand what the story means — it is the analysis the change acts on. It must not edit it. Core-file recall gets it into the prompt; the codegen allowlist keeps it out of the diff; and this target carries its own `_validate_enroldirect_file_set`, which fails loudly if a read-only file comes back modified.

Why this target is different from the other two

Its baseline is a removal. The checked-in source is the state *after* the impact analysis and *before* the gate acts on it. Nothing is broken; something is missing, and the missing thing has already been sized. The pristine copy lives in `.baseline/` and `demo/reset_s3_enroldirect.sh` restores it by copying that snapshot back.

The seeded-bug beat

```
rel_path: repos/enroldirect/eligibility.py
old: if category == PROSPECT:
new: if category == GUEST:
```

Redirects the prospect branch to match guests, so prospects fall through to no preference and are refused again — silently reverting the story at the gate. The generated tests are expected to catch it. Re-verify the snippet after any re-record: one that stops matching *fails silently*, and the "the tests caught it" claim never gets made.

11 · What proves it — and what does not

The regression suite is the AI's blind spot on purpose

`tests/test_regression_enroldirect.py` is checked in, human-authored, and named by **no** testgen or codegen allowlist — `tests/test_s3_testrun.py` asserts that. It lives outside the target root because anything ending `.py` under one joins the codegen candidate pool, and a suite the pipeline can rewrite is not an independent check of anything.

Every assertion holds before and after the story. It pins the ground that must not move, not the answer under debate:

IT ASSERTS	BECAUSE
The contract gate runs before the preference gate	Reversing them lets stale configuration on a dead contract grant access, and every category-level assertion would still pass.
Members' and guests' outcomes, named applicant by applicant, as <i>absolute</i> values	A comparison of two things the story changes together proves nothing.
The gate matches the contract configuration for classified categories	Pins <code>impact.py</code> 's model to the rules it claims to mirror, so the two cannot drift in either direction.
The analysis still reports disagreement <i>in both directions</i>	One column, or none, and the comparison feeding the recommendation has quietly stopped working.
Only one other team is owed work	Nobody notices extra tickets are fictional until three teams have triaged them.
Nothing the gate refuses can be enrolled, by any route	The enrolment path must not grow its own copy of the access rules.

Assertions go through HTTP and read fields off JSON rather than constructing decisions directly, so *adding* a field to the decision payload — which the story does — is not a regression. Nothing asserts the absence of a field.

Suggested run order on stage

1. Overview — 6 prospects, the two charts, the recommendation.
2. Access check — AP-4001 granted, then **AP-4005 denied**. Say the "not a decision" line.
3. Enrol as AP-4005 — channel closed, catalogue visible and untouchable.
4. Preference analysis → Consumers — one team is owed work: DocumentHub.
5. Run the pipeline on US-2026-045, apply, restart.
6. Back to Access check, AP-4005 — **granted under Guest access**.

Apply restarts the app, and that is the whole point

Applying rewrites the target's `.py` files, but a running uvicorn keeps serving the code it imported at startup. Without a restart the console says "the app now has this capability" while `:8083` still answers with the baseline — which reads to the audience as *"the change did nothing"*.

VERIFY ON THE APP, NOT IN THE CONSOLE

After Apply, re-run the **Access check** for AP-4005 Marguerite Vasseur. Granted, authorised by `Online Enrolment - Guest`, with the policy recorded — that is the beat. A failed restart is reported as such and never dressed up as a green banner.

What this walkthrough does not evidence

- **The figures are the seed's, not the book's.** 6 prospects, 3 disagreements, 2 packs are counts over twelve synthetic applicants, sized to make the comparison demonstrable — not an estimate of any real population.
- **The consumer inventory is declared, not discovered.** It is the analysis's stated input, and wrong the moment a consumer appears without the table being updated.
- **The DocumentHub figure assumes** it still selects wording on the authorising preference alone — taken from the integration contract, and the one thing here that could be out of date silently.
- **The source-control flow is modelled.** Branch → commit → push is rendered as a transcript; nothing runs git, and the release record puts the un-run pipeline in its "Not evidenced by this release" block so a modelled push can never read as a deployment that happened.